Preview of the subsection "Tolerable Current Against Emission Limits"

2.5 MVA, 20/0.4 kV unit, rated load, $\sigma = 5\%$, $\varepsilon = 0.3$, $c = 5.33$, 25 °C ambient unless stated. Numbers from limits_study.py, 29 Oct 2026.

A Tolerable Current Against Emission Limits

Fig. 1 sets the tolerable current I_h^* against the per-order limits of IEEE 519 for $I_{SC}/I_L < 20$, the row that binds generation facilities [1, Table 2], and against the 5% total rated-current distortion (TRD) of IEEE 1547, the only limit that reaches above the 50th order [2, Sec. 7.3]. Both are referred to the rated current of the transformer, so the converter or load is taken to be rated at the transformer rating, and the 0.3% limit of the 35th to 50th orders is extended upward as a reference that no standard sets. Table I lists where each case meets these limits.

Up to the 50th order no case reaches a limit. At rated load the limits lie at least 3.3 times below I_h^*, the tightest margin falls at the 9th and 10th orders at the top of the 4% band, and it hardly depends on the eddy-loss model. A stray fraction of 15% narrows the margin to 2.1, close to the multiplier of up to 2.2 that IEEE 519 allows when lower orders are suppressed [1, Table 5], while part load widens it to 8 to 11. Below 2.5 kHz the existing limits therefore protect the transformer.

Above the 50th order only the TRD remains, and a converter that meets it by concentrating its distortion in a single order exceeds I_h^* at rated load and the nominal stray fraction from the 51st order onward for every model and thickness except the diffusion solution for 4 mm, where the excess begins at 3.2 kHz. At 10 kHz the excess is 6.5 under the square law, 4.9 to 6.5 under Annex C and 1.3 to 6.1 under the diffusion solution for 1 to 4 mm, and it grows with frequency to as much as 16 at 25 kHz. The stray fraction moves the square-law excess at 10 kHz between 5.1 and 11.1, and only the thickest conductor with the smallest stray fraction brings the diffusion solution down to unity.

The extended 0.3% line shows where a per-order limit would have to lie. At the nominal stray fraction no boundary falls below it up to 25 kHz, the square law ending at 0.31%, whereas a stray fraction of 15% takes the square law below it from 15 kHz and Annex C for 1 and 2.5 mm and the diffusion solution for 1 mm from 15 to 19 kHz. The transformer is thus protected by a per-order level close to the 0.3% that IEEE 519 already sets at the 35th to 50th orders, but not by the 5% that the TRD leaves to a single kilohertz order.

Part load widens the thermal headroom but does not close the gap. At 0.75 pu a TRD-compliant converter still exceeds I_h^* from 3.9 kHz under the square law and from 8.4 kHz under the diffusion solution for 2.5 mm, and at 0.5 pu the square law and Annex C are exceeded from about 5 kHz, so that only the diffusion solution at half load keeps I_h^* above the TRD up to 25 kHz.

Table I. Tolerable Current Against the Emission Limits

Case	Margin to 519	Below 0.3% from [kHz]	TRD exceeds from [kHz]	TRD$/I_h^*$ at 10 kHz
h^2	3.3	–	2.55	6.5
Annex C, 2.5 mm	3.3	–	2.55	6.0
Diffusion, 1 mm	3.3	–	2.55	6.1
Diffusion, 2.5 mm	3.4	–	2.55	2.6
Diffusion, 4 mm	3.5	–	3.2	1.3
h^2, $\sigma = 15\%$	2.1	15.1	2.55	11.1
h^2, 0.5 pu	10.7	–	4.95	2.0
Diffusion, 2.5 mm, 0.5 pu	10.8	–	–	0.8

Rated load and $\sigma = 5\%$ unless stated. Margin is the smallest ratio of I_h^* to the IEEE 519 limit up to $h = 50$, and – means not reached up to 25 kHz.

References

[1] *IEEE Standard for Harmonic Control in Electric Power Systems*, IEEE Std 519-2022, 2022.

[2] *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces*, IEEE Std 1547-2018, 2018.

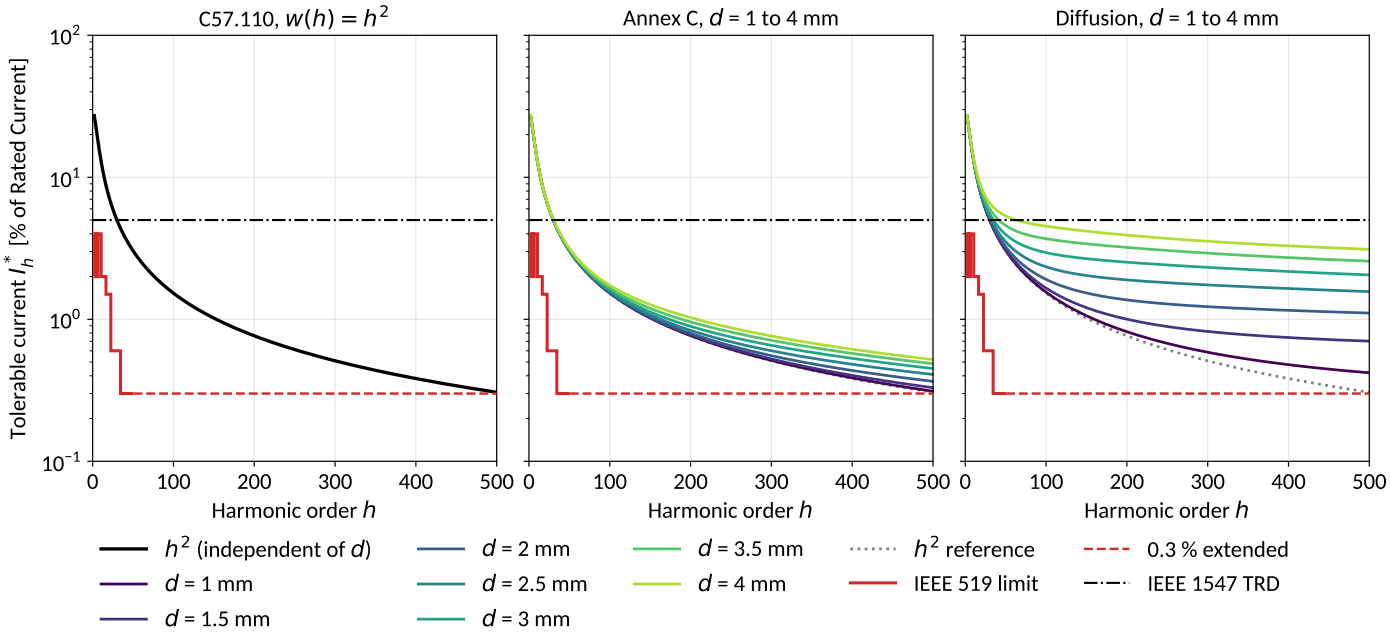


Figure 1. Tolerable current I_h^* at rated load under the C57.110 square law (left), its Annex C correction (middle) and the diffusion solution (right), the latter two for conductor thicknesses of 1 to 4 mm, against the IEEE 519 per-order limits for $I_{SC}/I_L < 20$, the 0.3% limit extended above the 50th order and the 5% TRD of IEEE 1547. A harmonic above a curve ages the insulation faster than normal.